

Infrastructure Module Reference

Infrastructure Module Reference I

This section inventories every Layer-1 subdirectory returned by `28 discover_infrastructure_modules(repo_root)`. File totals use 708 Python sources across infra + 9,557 infra tests guarding them. Documentation Duality = paired `README.md` + `AGENTS.md`; optional `SKILL.md` manifests feed `python -m infrastructure.skills`.

Module	Python Files	Has AGENTS.md	Has README.md	Key Exports
autoresearch	10			<code>build_autoresearch_plan</code> , <code>readiness validation CLI</code>
benchmark	4			Template harness scoring + comparative gates
config	0			Repository defaults + hardened templates
core	120			<code>get_logger</code> , <code>load_config</code> , <code>TemplateError</code>
docker	0			Containerisation scaffolding
doctor	14			Checkout <code>diagnose/fix/undo repairs</code>
documentation	14			<code>FigureManager</code> , <code>generate_glossary</code>
fonds	6			—

Infrastructure Module Reference II

llm	55	Ollama helpers, sanitization, review + translation pipelines
logrotate.d	0	Rotation snippets (documentation-first)
methods	5	build_methods_orche stration_plan, methods-stage contracts + validation
orchestration	12	PipelineRunner, entry point for ./run.sh
project	41	discover_projects, workspace management
prose	9	Markdown readability + prose tooling
provenance	7	—
publishing	81	Zenodo, executable bundle, archival targets
reference	16	BibTeX models, parsers, converters
rendering	65	PDF/HTML/slide rendering, Pandoc filters
reporting	57	Coverage parsers, dashboards, executive artefacts
research	3	—
rules	6	—
scientific	4	check_numerical_sta bility, benchmark_f unction
search	62	infrastructure.sear ch.literature clients + cache

Infrastructure Module Reference III

sia	10	Self-Improving-AI loop: task validation, harness, metric capture
skills	8	discover_skills, SKILL manifest regeneration
steganography	13	Watermark overlays + hash manifests
tools	6	—
validation	80	PDF + Markdown + integrity CLIs

Alphabetical summaries I

Below, `${module}_*_python_file_count` placeholders expand per subdirectory at render-time.

`infrastructure.autoresearch` (10 files)

Readiness planner, validation CLI, and report models for AutoResearch-style project promotion (`infrastructure/autoresearch/`).

`infrastructure.benchmark` (4 files)

Template harness scoring and comparative gate helpers exercised in CI smoke paths.

`infrastructure/config` (non-package subdirectory)

Repository-wide YAML templates and secure manifests (`.env.template`, hardened defaults referenced by Docker + CLI). `config/` carries no `__init__.py`, so it is a configuration subdirectory rather than an importable package.

Alphabetical summaries II

`infrastructure.core` (120 files)

Checkpointing, logging, pipeline YAML parsing, telemetry bridges, filesystem helpers, hardened exceptions. Everything else imports logging + error taxonomy from here first.

`infrastructure.doctor` (14 files)

Checkout diagnose/fix/undo repairs for broken local workspace states.

`infrastructure.docker` (0 files)

Pinned images / compose scaffolding for reproducible CI + remote builds.

`infrastructure.documentation` (14 files)

Figure registries plus glossary tooling feeding manuscript automation.

Alphabetical summaries III

`infrastructure.fonds` (6 files)

Resource pool management for curated fonds (tracked reference datasets, bibliographic collections, and evidence corpora). Fonds mirror `projects/templates/` with `git-tracked templates/*` exemplars and sidecar-linked private lifecycle folders.

`infrastructure.llm` (55 files)

Ollama integrations, sanitization adapters, templated reviewer flows. **Literature ingestion now lives primarily in `search/literature` + citation helpers in `reference/`.**

`infrastructure.methods` (5 files)

Deterministic methods-orchestration contracts (`MethodStage`, `MethodsOrchestrationPlan`, `MethodsIssue`): builds and validates an ordered methods plan for a research project so the manuscript's "Methods" track stays bound to executable stages.

Alphabetical summaries IV

`infrastructure.orchestration` (12 files)

`python -m infrastructure.orchestration` exposes interactive menus, subprocess wiring for thin shell wrappers (`run.sh`, `secure_run.sh`), and stubs used in CI for menu parsing tests.

`infrastructure.project` (41 files)

Canonical discovery (`discover_projects`) enforcing `src/` + `tests/`, slug validation, nested WIP namespaces.

`infrastructure.prose` (9 files)

Readability metrics + Markdown tooling for prose-centric manuscripts / CI gates.

Alphabetical summaries V

`infrastructure.provenance` (7 files)

Content-addressed provenance DAG. Records artifact lineage (which run produced which file, from which inputs) as a verifiable graph of artifact/run/source/claim nodes connected by produced/consumed/derived-from/supports/refutes edges. Includes a structured Review system with severity (blocking/major/minor/info) and verdict (refutes/supports). Features a CLI and pipeline integration hooks for automatic lineage recording after every stage.

`infrastructure.publishing` (81 files)

Metadata models, APA/BibTeX/MLA formatters, optional Zenodo clients.

`infrastructure.reference` (16 files)

Citation/BibTeX parsing + conversion utilities leveraged by manuscripts and retrieval scripts.

Alphabetical summaries VI

`infrastructure.rendering` (65 files)

Pandoc shim, Unicode/XeLaTeX postprocessors, combined PDF/HTML/slide exporters.

`infrastructure.reporting` (57 files)

Parses pytest + coverage artefacts for dashboards; pairs with Stage 01 summaries and downstream executive exports.

`infrastructure.research` (3 files)

Seven-stage research workflow

(SCOPE→LITERATURE→REASON→DESIGN→COMPUTE→SYNTHE

defined as typed ResearchStage data classes with explicit outputs, skills, and template commands per stage. Includes a full PRISMA-adapted literature review prompt and a research workflow prompt referencing template/ infrastructure commands.

Alphabetical summaries VII

`infrastructure.rules` (6 files)

Governance rules layer for validating project lifecycle transitions, sidecar sync policies, and pipeline gate orchestration. Rules mirror `projects/templates/` with git-tracked `templates/*` exemplars.

`infrastructure.scientific` (4 files)

Stability probing, benchmarking hooks—consumed heavily by optimization exemplars (`template_code_project` scripts).

`infrastructure.search` (62 files)

Two-tier search architecture: the `literature/` client stack (`client.py`, `backends`, `caches`) powers literature search with arXiv, Crossref, local, and Paperclip backends. `connectors/` exposes the built-in scientific database adapters through a uniform `Connector Registry`; OpenAlex, UniProt, PDB, Semantic Scholar, European PMC, bioRxiv, and other registered adapters share `list/search` CLI commands with HTTP timeout, retry, and TTL caching. The live registry, not this prose, is authoritative for connector count.

Alphabetical summaries VIII

`infrastructure.sia` (10 files)

Generic Self-Improving-AI loop utilities: task-layout validation, execution harness, and metric capture reused by `template_sia` (fixture-replay by default).

`infrastructure.skills` (8 files)

Discovers SKILL.md frontmatter → `.cursor/skill_manifest.json`.

`infrastructure.steganography` (13 files)

Watermark overlays, hashing companions triggered by secure pipeline path.

`infrastructure.tools` (6 files)

Invocable tool definitions registered by resource-pool governance; tools mirror `projects/templates/` with git-tracked `templates/*` exemplars.

Alphabetical summaries IX

`infrastructure.validation` (80 files)

Markdown + PDF + integrity CLIs underpinning Stage 04 diagnostics.

`infrastructure/logrotate.d` (0 files)

Operational templates for deployments (documentation-first; intentionally minimal Python footprint).

Documentation maturity: Coverage statements in Results pull from introspection—not hand-maintained denominators—so newly promoted modules automatically flow into manuscripts after `generate_manuscript_metrics.py`.

FAIR+RSE linkage: MCP-ready SKILL.md artefacts align with evaluator heuristics (executability + metadata richness) emphasized by FAIRsoft guidance [[@garijo2024fairsoft](#)].